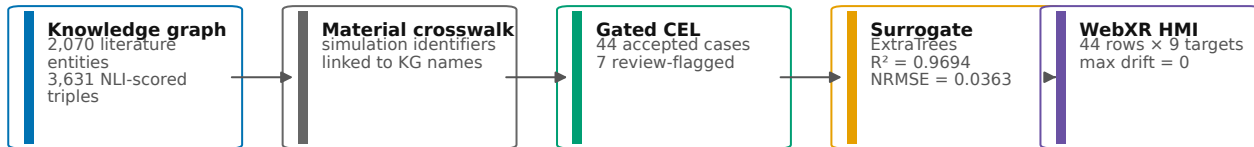


a

The virtual twin is treated as an auditable decision-support chain, not as an autonomous process controller.

b

Authorized outputs

- qualified-pair interpolation of Abaqus/Explicit CEL response quantities
- browser-executable surrogate inference with exact Python-runtime replay
- KG-backed provenance and operator-context visualization

Blocked claims

- ✗ experimental physical validation claims
- ✗ bond/no-bond threshold claims
- ✗ unseen material-pair prediction
- ✗ autonomous process control